

Computer Networks: Assignment-9

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Q 1) Configure a wireless connection as shown in the below network.

Diagram:



Router Config:

Wireless Settings:

Basic Wireless Settings:

Network Name: MyWiFi

Wireless Security:

Security Mode: WEP

Key1: 1234567890

IP Config:

PC0: 192.168.0.5 / 255.255.255.0 / 192.168.0.1

PC1: 192.168.0.6 / 255.255.255.0 / 192.168.0.1

PC2: 192.168.0.7 / 255.255.255.0 / 192.168.0.1

PC Wireless:

Connect:

Link Information

Connect

Profiles

Below is a list of available wireless networks. To search for more wireless networks, click the **Refresh** button. To view more information about a network, select the wireless network name. To connect to that network, click the **Connect** button below.

Wireless Network Name	CH	Signal
MyWiFi	1	100%

Site Information

Wireless Mode Infrastructure
Network Type Mixed B/G/N
Radio Band Auto
Security WEP
MAC Address 000C:CF02:6B06

Refresh

Connect

2.4GHz



Adapter is Active

Wireless-N Notebook Adapter Wireless Network Monitor v1.0 Model No. WPC300N

Output:

```
PC0
Physical  Config  Desktop  Programming  Attributes
Command Prompt
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 39ms, Maximum = 69ms, Average = 54ms

C:\>ping 192.168.0.6

Pinging 192.168.0.6 with 32 bytes of data:

Reply from 192.168.0.6: bytes=32 time=51ms TTL=128
Reply from 192.168.0.6: bytes=32 time=39ms TTL=128
Reply from 192.168.0.6: bytes=32 time=69ms TTL=128
Reply from 192.168.0.6: bytes=32 time=58ms TTL=128

Ping statistics for 192.168.0.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 39ms, Maximum = 69ms, Average = 54ms

C:\>ping 192.168.0.1

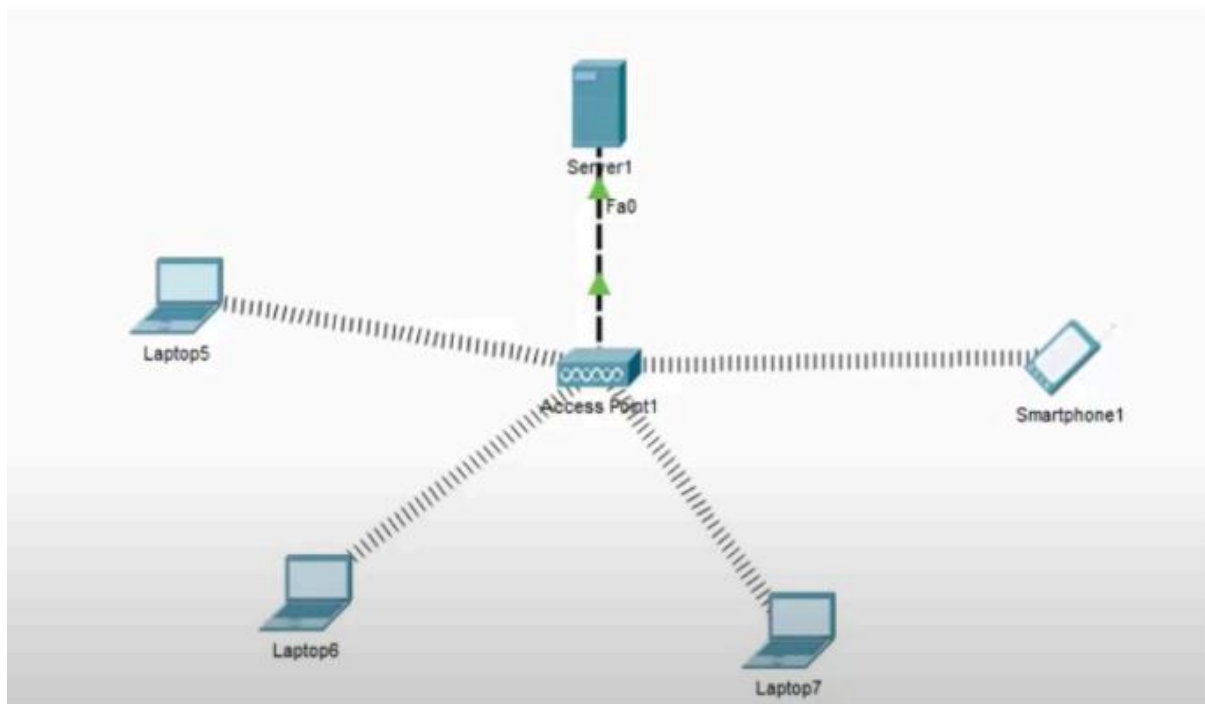
Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time=28ms TTL=255
Reply from 192.168.0.1: bytes=32 time=15ms TTL=255
Reply from 192.168.0.1: bytes=32 time=23ms TTL=255
Reply from 192.168.0.1: bytes=32 time=31ms TTL=255

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 15ms, Maximum = 31ms, Average = 24ms
```

Q2) Configure the below network using a server and connect wirelessly all the device.

Diagram:



IP Config:

Server0: 172.16.0.1 / 255.255.255.0

Server0 DHCP Config:

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
WIFI	172.16.0.1	172.16.0.1	172.16.0.2	255.255.2...	254	0.0.0.0	0.0.0.0

Received IPs:

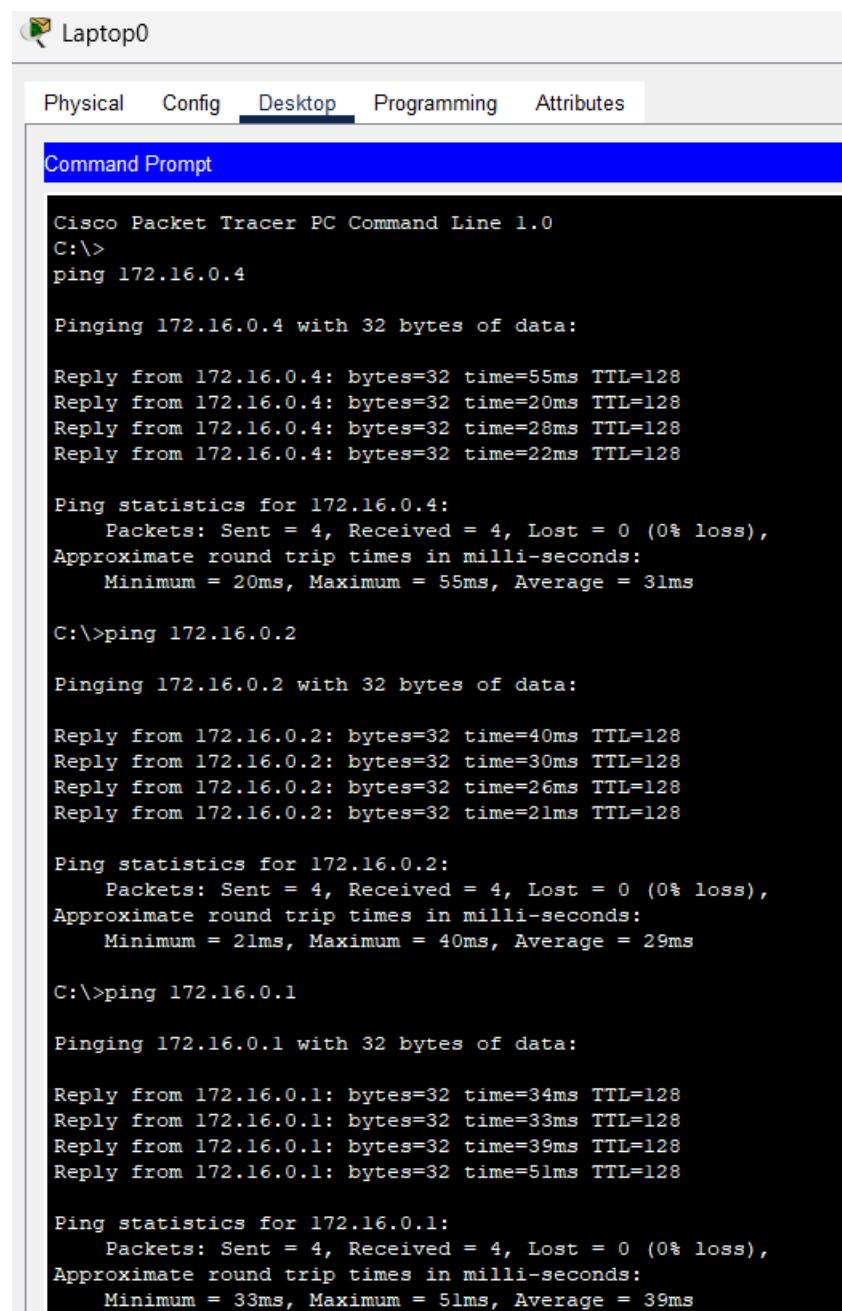
Laptop0: 172.16.0.8 / 255.255.255.0

Laptop1: 172.16.0.7 / 255.255.255.0

Laptop2: 172.16.0.5 / 255.255.255.0

Smartphone0: 172.16.0.2 / 255.255.255.0

Output:



The screenshot shows the Cisco Packet Tracer PC Command Line interface for a device named Laptop0. The interface has tabs for Physical, Config, Desktop, Programming, and Attributes, with Desktop selected. The Command Prompt window displays the results of three ping commands: ping 172.16.0.4, ping 172.16.0.2, and ping 172.16.0.1. Each command shows four replies with their respective byte sizes, times, and TTL values, followed by a summary of the ping statistics.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>
ping 172.16.0.4

Pinging 172.16.0.4 with 32 bytes of data:

Reply from 172.16.0.4: bytes=32 time=55ms TTL=128
Reply from 172.16.0.4: bytes=32 time=20ms TTL=128
Reply from 172.16.0.4: bytes=32 time=28ms TTL=128
Reply from 172.16.0.4: bytes=32 time=22ms TTL=128

Ping statistics for 172.16.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 20ms, Maximum = 55ms, Average = 31ms

C:\>ping 172.16.0.2

Pinging 172.16.0.2 with 32 bytes of data:

Reply from 172.16.0.2: bytes=32 time=40ms TTL=128
Reply from 172.16.0.2: bytes=32 time=30ms TTL=128
Reply from 172.16.0.2: bytes=32 time=26ms TTL=128
Reply from 172.16.0.2: bytes=32 time=21ms TTL=128

Ping statistics for 172.16.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 21ms, Maximum = 40ms, Average = 29ms

C:\>ping 172.16.0.1

Pinging 172.16.0.1 with 32 bytes of data:

Reply from 172.16.0.1: bytes=32 time=34ms TTL=128
Reply from 172.16.0.1: bytes=32 time=33ms TTL=128
Reply from 172.16.0.1: bytes=32 time=39ms TTL=128
Reply from 172.16.0.1: bytes=32 time=51ms TTL=128

Ping statistics for 172.16.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 33ms, Maximum = 51ms, Average = 39ms
```